Our problem statement:
Are meteorological, agricultural, and satellite features (i.e. from the CY-Bench Dataset) good predictors of European Wheat Futures Price Changes?
Research Report Structure:

<u>Advice from the formative and summative project briefs:</u>
- The report should be structured in the way that best fits the story we're telling
    o Intro (background info on the dataset & prediction problem at hand) → <u>define context and scope</u>
    o Brief exploratory data analysis → give information you took out form exploring the dataset's data that is important in you pursuing the task (or is of explicit importance)
    o Information about all the aspects of our analysis (Bulk of the report): data preprocessing → quantitative results → model comparison
    o Conclusion discussing: Learnings, ideas for further improvement, etc.
- Literature review can be included if relevant to our Dataset
 	
<u>Past MA429 Projects:</u>
- https://github.com/ZXChen299/Previous-LSE-MA429-Projects/blob/master/MA429%20Final%20Project/IMDB/Report.pdf
    o **Note: the report is overall not great**
    o Structure of main body: Intro (subsections: Motivation, Goal and Limitations, Structure of the report), Data Pre-processing (subsections: preparation of the dataset, exploratory analysis on the training set), Performance Measures (subsections: regression, classification), Regression (subsections: simple and multiple regression, estimating and interpreting model coefficients, issues and improvements to the regression model, making predictions from the linear model), Classification (subsections: model tuning and training, application and performance on testing dataset), Conclusion (subsections: interpretation of results)
- No others were found
 	
<u>From research:</u>
- Advice from Carnegie Mellon: https://www.stat.cmu.edu/~brian/701/notes/paper-structure.pdf
    o Don't use complex language
    o Appendix = more detailed and process-oriented
    o Proposed structure:
        ▪ **Intro containing =** summary of the study and data + any relevant substantive context / background / framing issues
        ▪ The Big Questions answered by our data analysis and a summary of our conclusions about these questions
        ▪ Brief outline of the remainder of the paper
        ▪ Body = can be traditional (data/methods/analysis/results) or question oriented (i.e. one section per question raised in the introduction)

General Research Report Advice:
- 16 pages limit (excluding title page, executive summary, table of contents, bibliography and appendices) - No appendix page limit but only include relevant things / things of interest

Proposed Main Body Structure

- **Introduction =** data set key info, chosen prediction task and justification, context and scope of the task, summary of the study conducted and results (1 page)
- **(Optional) Brief Literature Review =** review some of the key papers on the dataset as well as the prediction task at hand (could help set expectations for the reader) – (1 page)
- **Data Pre-processing =** Details of how we managed the regional data split when fitting our models (maybe: method 1 = fit one model per region, method 2 = aggregation) (2 pages)
- **Redefining the prediction task =** important section to explain how we change the prediction task → here we're addressing questions such as: why are we predicting 5 day ahead returns and not daily returns? How are we dealing with the futures rolling cost in our time series price? → this should be brief and clear (1 page)
- **Feature engineering and selection =** How we augmented our data to add predictive power, using insights from an agricultural (e.g. if temperature is consistently high for a period and precipitation is low, this may indicate a drought) | also we should indicate how we performed feature selection before fitting all of our models (e.g. using a decision tree and looking at feature importance / e.g. putting features on the same scale and looking at their correlation) (2 pages)
- **For each method used =**
    - Method justification = why we believe this method would improve predictive power
    - Analysis = how we tuned the method, our methodology and justification
    - Results = what was the predictive power of this method

  (4 times 2 pages)

- **Model Results Comparison =** Comparing and interpreting results across models, proposing explanations for outcomes (1 page)
- **Conclusion =** Discussing results, drawing some key learnings from the discussion, ideas for further improvement, proposed future work (1 page)

Formative Feedback:
General: - Writing was mostly clear but could be improved - Instructions were not followed well (no gen AI statement, names should not have been included, code must be submitted separately) Context, modelling and originality - The consideration of learning methods + imputation methods together came out as a little confusing Analysis - Comparison was hard to follow and could have been structured better (Neil said: "having some tables comparing the most important metrics, and then discussing these, would be easier to follow I think") - Check for errors in the analysis (e.g. when implementing methods) → there were 2 errors in the analysis in the summative Organisation and presentations - Two-column format is STRONGLY discouraged - Neil: "I suggest having headline results in a table in the main text (leaving more in-depth / less important tables in the appendix)" - Avoid mini lecturing - Inconsistent citation formatting

- Bibliography should be before the appendix not after. - We made a good use of the Appendix during the summative Code Quality - Code was nicely structured - Table of Contents is helpful - Hard coded grid search means results can't be verified - Variable naming inconsistencies (train_df vs. df_train)
➔ No specific changes are made to the structure according to the formative feedback – the advice should be kept in mind when writing and coding
Comments directly inside the report: - Make sure that your objective and goal are clear in the introduction, especially when no context has yet been specified - Making a list of the feature names without explaining what they are / their meaning to the reader is not useful - Make sure that references bring something clear to your argument → that was a key comment in some of the paragraphs of the formative assessment - Avoid wordy sentences → going more straight to the point is better - Make sure that inexplicit terms are explained / defined. - <u>AVOID:</u> mini-lectures of basic stuff that your target reader should know and that we don't have the space to explain even if they don't